1 Two-loop Differential Collinear Kernel

We consider the double collinear building block. The two masses inside each collinear pair are kept distinct:

$$\mathcal{I}(x,y)=\int\frac{d^dk}{i\pi^{d/2}}\frac{d^dl}{i\pi^{d/2}}\,\frac{\delta(x-l\cdot\eta)\,\delta(y-k\cdot\eta)}{[l^2-m_l^2+i0][(l-p)^2-M_l^2+i0][k^2-m_k^2+i0][(l+k-p)^2-M_k^2+i0]}. \tag{1}$$

We normalize

$$p\cdot\eta=1,\qquad \eta^2=0. \tag{2}$$

Define

$$K(p-l;y)\equiv\int\frac{d^dk}{i\pi^{d/2}}\,\frac{\delta(y-k\cdot\eta)}{[k^2-m_k^2+i0][(k-p+l)^2-M_k^2+i0]}. \tag{3}$$

Then

$$\mathcal{I}(x,y)=\int\frac{d^dl}{i\pi^{d/2}}\,\frac{\delta(x-l\cdot\eta)}{[l^2-m_l^2+i0][(l-p)^2-M_l^2+i0]}\,K(p-l;y). \tag{4}$$

1.0.1 Light-cone decomposition

$$l=xp+\beta_l\eta+l_\perp,\qquad k=yp+\beta_k\eta+k_\perp, \tag{5}$$

with

$$p\cdot l_\perp=\eta\cdot l_\perp=0,\qquad p\cdot k_\perp=\eta\cdot k_\perp=0. \tag{6}$$

Then indeed

$$l\cdot\eta=x,\qquad k\cdot\eta=y. \tag{7}$$

In these variables,

$$d^dl=dx\,d\beta_l\,d^{d-2}l_\perp,\qquad d^dk=dy\,d\beta_k\,d^{d-2}k_\perp. \tag{8}$$

1.0.2 The two k-dependent denominators

The first denominator is

$$k^2=y^2p^2+2y\beta_k+k_\perp^2,\qquad D_1\equiv k^2-m_k^2+i0=2y\beta_k+\big(y^2p^2+k_\perp^2-m_k^2\big)+i0. \tag{9}$$

For the second denominator we use

$$k-p+l=(x+y-1)p+(\beta_k+\beta_l)\eta+(k_\perp+l_\perp),$$

so that

$$(k-p+l)^2=(x+y-1)^2p^2+2(x+y-1)(\beta_k+\beta_l)+(k_\perp+l_\perp)^2, \tag{10}$$

and hence

$$D_2\equiv(k-p+l)^2-M_k^2+i0=2(x+y-1)(\beta_k+\beta_l)+\big((x+y-1)^2p^2+(k_\perp+l_\perp)^2-M_k^2\big)+i0. \tag{11}$$

Equivalently, with

$$q=p-l=(1-x)p-\beta_l\eta-l_\perp,\qquad q\cdot\eta=1-x,$$

one may write

$$D_2=2\big(y-(1-x)\big)(\beta_k+\beta_l)+\big((y-(1-x))^2p^2+(k_\perp+l_\perp)^2-M_k^2\big)+i0. \tag{12}$$

Thus

$$K(p-l;y)=\int\frac{d\beta_k\,d^{d-2}k_\perp}{i\pi^{d/2}}\,\frac{1}{D_1D_2}. \tag{13}$$

1.0.3 Pole structure in β_k and support in y

The two poles in the complex β_k plane are

$$\beta_k^{(1)} = \frac{m_k^2 - k_\perp^2 - y^2p^2 - i0}{2y}, \tag{14}$$

and

$$\beta_k^{(2)} = -\beta_l + \frac{M_k^2 - (k_\perp + l_\perp)^2 - (y-(1-x))^2p^2 - i0}{2(y-(1-x))}. \tag{15}$$

The β_k integral is nonzero only when the two poles lie on opposite sides of the real axis. Assuming $1-x>0$, this happens precisely for

$$0 < y < 1-x. \tag{16}$$

Therefore the subloop carries the support

$$\Theta(0 \le y \le 1-x). \tag{17}$$

1.0.4 The β_k integral by residues

We evaluate the β_k integral by taking the residue at the pole of D_1. Since the pole is simple,

$$\mathrm{Res}_{\beta_k=\beta_k^{(1)}} \frac{1}{D_1D_2} = \frac{1}{D_1'(\beta_k^{(1)})\, D_2(\beta_k^{(1)})}, \qquad D_1'(\beta_k) = 2y.$$

Hence the residue contributes the factor $1/(2y)$.
At the pole $\beta_k = \beta_k^{(1)}$ one finds

$$\begin{aligned} D_2(\beta_k^{(1)}) = {} & \frac{y-(1-x)}{y}(m_k^2 - k_\perp^2 - y^2p^2) + 2(y-(1-x))\beta_l \\ & + (y-(1-x))^2p^2 + (k_\perp + l_\perp)^2 - M_k^2 + i0. \end{aligned} \tag{18}$$

Using $(k_\perp + l_\perp)^2 = k_\perp^2 + 2k_\perp\cdot l_\perp + l_\perp^2$, this becomes

$$\begin{aligned} D_2(\beta_k^{(1)}) = {} & -\frac{1-x}{y}k_\perp^2 + 2k_\perp\cdot l_\perp \\ & + \left[\frac{y-(1-x)}{y}m_k^2 + 2(y-(1-x))\beta_l + l_\perp^2 - (1-x)(x+y-1)p^2 - M_k^2\right] + i0. \end{aligned} \tag{19}$$

Next we use the exact identity

$$(p-l)^2 = (1-x)^2p^2 - 2(1-x)\beta_l + l_\perp^2, \tag{20}$$

which implies

$$l_\perp^2 = (p-l)^2 - (1-x)^2p^2 + 2(1-x)\beta_l.$$

After a short algebraic rearrangement, it is convenient to introduce

$$\lambda \equiv \frac{y}{1-x}, \qquad \mu_k^2(\lambda) \equiv (1-\lambda)m_k^2 + \lambda M_k^2. \tag{21}$$

and one obtains

$$D_2(\beta_k^{(1)}) = -\frac{1-x}{y}\left[(k_\perp - \lambda l_\perp)^2 + \Delta_\perp - i0\right], \tag{22}$$

where

$$\Delta_\perp = \mu_k^2(\lambda) - \lambda(1-\lambda)(p-l)^2 - \lambda(1-\lambda x)p^2, \tag{23}$$

Therefore

$$\frac{1}{D_2(\beta_k^{(1)})} = -\frac{y}{1-x}\frac{1}{(k_\perp - \lambda l_\perp)^2 + \Delta_\perp - i0}. \tag{24}$$

Combining this with the residue factor $1/(2y)$, the factors of y cancel and one gets

$$K(p-l;y) = \frac{\Theta(0 \le y \le 1-x)}{1-x}\int \frac{d^{d-2}k_\perp}{\pi^{(d-2)/2}}\frac{1}{(k_\perp - \lambda l_\perp)^2 + \Delta_\perp - i0}. \tag{25}$$

Now shift

$$r_\perp = k_\perp - \lambda l_\perp, \qquad d^{d-2}k_\perp = d^{d-2}r_\perp,$$

and use the standard Euclidean integral in $n = d-2 = 2-2\epsilon$ dimensions,

$$\int \frac{d^n r}{\pi^{n/2}}\frac{1}{(r^2+\Delta)^\alpha} = \frac{\Gamma(\alpha - n/2)}{\Gamma(\alpha)}\,\Delta^{n/2-\alpha}.$$

With $\alpha = 1$ this gives

$$K(p-l;y) = \Gamma(\epsilon)\frac{\Theta(0 \le y \le 1-x)}{1-x}\Big[\mu_k^2(\lambda) - \lambda(1-\lambda)(p-l)^2 - \lambda(1-\lambda x)p^2 - i0\Big]^{-\epsilon}. \tag{26}$$

Equivalently,

$$K(p-l;y) = \Gamma(\epsilon)\frac{\Theta(0 \le y \le 1-x)}{1-x}\,[C(l) - \mathrm{i}0]^{-\epsilon}, \tag{27}$$

where

$$C(l) = \mu_k^2(\lambda) - \lambda(1-\lambda)(p-l)^2 - \lambda(1-\lambda x)p^2. \tag{28}$$

Thus the full integral is reduced to

$$\boxed{\mathcal{I}(x,y) = \Gamma(\epsilon)\frac{\Theta(0 \le y \le 1-x)}{1-x}\int \frac{\mathrm{d}^d l}{\mathrm{i}\pi^{d/2}}\frac{\delta(x - l\cdot\eta)}{[l^2 - m_l^2 + \mathrm{i}0][(l-p)^2 - M_l^2 + \mathrm{i}0]}[C(l) - \mathrm{i}0]^{-\epsilon}.} \tag{29}$$

1.1 Method 1: Simultaneous Parametrization

1.1.1 Combine all l-dependent factors simultaneously

Introduce

$$A = l^2 - m_l^2 + i0, \qquad B = (l-p)^2 - M_l^2 + i0. \tag{30}$$

Then

$$\frac{\Gamma(\epsilon)}{A\,B\,C^\epsilon} = \Gamma(2+\epsilon)\int du\,dv\,dw\,\delta(1-u-v-w)\frac{w^{\epsilon-1}}{[uA+vB+wC]^{2+\epsilon}}. \tag{31}$$

Hence

$$\mathcal{I}(x,y) = \frac{\Gamma(2+\epsilon)\Theta(0\le y\le 1-x)}{1-x}\int_{\Delta_2} du\,dv\,dw\,w^{\epsilon-1}\int\frac{d^dl}{i\pi^{d/2}}\frac{\delta(x-l\cdot\eta)}{D(l)^{2+\epsilon}}, \tag{32}$$

where $u,v,w\ge 0$, $u+v+w=1$ and

$$D(l) = uA + vB + wC. \tag{33}$$

1.1.2 Quadratic form in l

Writing

$$a = \lambda(1-\lambda), \qquad b = \lambda(1-\lambda x), \tag{34}$$

and

$$\mu_l^2(x) \equiv (1-x)m_l^2 + xM_l^2, \qquad \mu_l^2(u) \equiv um_l^2 + (1-u)M_l^2, \tag{35}$$

one finds after expansion

$$D(l) = Ul^2 - 2V\,l\cdot p + W + i0, \tag{36}$$

with

$$U = u + v - aw, \tag{37}$$

$$V = v - aw, \tag{38}$$

$$W = [v - w(a+b)]p^2 - um_l^2 - vM_l^2 + w\mu_k^2(\lambda). \tag{39}$$

Shift

$$l = q + \frac{V}{U}p,$$

so that

$$D(l) = Uq^2 + \Delta(u,v,w) + i0, \tag{40}$$

where

$$\Delta = W - \frac{V^2}{U}p^2. \tag{41}$$

1.1.3 Delta function by Fourier representation

Use

$$\delta(x-l\cdot\eta) = \int\frac{d\omega}{2\pi}e^{i\omega(x-l\cdot\eta)}.$$

Since $p\cdot\eta = 1$ and $\eta^2 = 0$, the Gaussian q-integration gives

$$\int\frac{d^dq}{i\pi^{d/2}}\frac{e^{-i\omega q\cdot\eta}}{(Uq^2+\Delta)^{2+\epsilon}} = U^{-d/2}\frac{\Gamma(2\epsilon)}{\Gamma(2+\epsilon)}\Delta^{-2\epsilon}, \tag{42}$$

independent of ω. The Fourier source in the shifted q-integral does not generate an ω-dependent factor because the source vector is lightlike. To see this, use a Schwinger representation,

$$\frac{1}{(Uq^2+\Delta)^{2+\epsilon}} = \frac{1}{\Gamma(2+\epsilon)}\int_0^\infty d\alpha\,\alpha^{1+\epsilon}e^{-\alpha(Uq^2+\Delta)}.$$

Then the q-dependent part is

$$\int d^d q \, \exp\Big[-\alpha U q^2 - i\omega q\cdot\eta\Big].$$

Completing the square gives the standard Gaussian result

$$\int d^d q \, \exp\Big[-\alpha U q^2 - i\omega q\cdot\eta\Big] = \left(\frac{\pi}{\alpha U}\right)^{d/2} \exp\left[-\frac{\omega^2\eta^2}{4\alpha U}\right].$$

Since the auxiliary vector is lightlike,

$$\eta^2 = 0,$$

the source-dependent exponential becomes

$$\exp\left[-\frac{\omega^2\eta^2}{4\alpha U}\right] = 1.$$

Therefore the Fourier source does not modify the Gaussian integral:

$$\int d^d q \, \exp\Big[-\alpha U q^2 - i\omega q\cdot\eta\Big] = \left(\frac{\pi}{\alpha U}\right)^{d/2}.$$

Consequently,

$$\int \frac{d^d q}{i\pi^{d/2}} \, \frac{e^{-i\omega q\cdot\eta}}{(Uq^2+\Delta)^{2+\epsilon}} = \int \frac{d^d q}{i\pi^{d/2}} \, \frac{1}{(Uq^2+\Delta)^{2+\epsilon}},$$

Hence

$$\int \frac{d^d q}{i\pi^{d/2}} \, \frac{e^{-i\omega q\cdot\eta}}{(Uq^2+\Delta)^{2+\epsilon}} = U^{-d/2} \frac{\Gamma(2\epsilon)}{\Gamma(2+\epsilon)} \Delta^{-2\epsilon}.$$

The essential point is that the source is proportional to η^μ, and therefore its square is proportional to η^2. Since $\eta^2 = 0$, the Fourier source leaves no ω-dependence behind.Therefore the ω integral returns

$$\delta\left(x - \frac{V}{U}\right).$$

Thus

$$\mathcal{I}(x,y) = \frac{\Gamma(2\epsilon)\Theta(0 \le y \le 1-x)}{1-x} \int_{\Delta_2} du\, dv\, dw\, w^{\epsilon-1} U^{-d/2} \delta\left(x - \frac{V}{U}\right) \Delta^{-2\epsilon}. \tag{43}$$

1.1.4 Collapse to one parameter

Using $u + v + w = 1$, one has

$$U = 1 - (1+a)w,$$

which is independent of v. The delta functions fix

$$v = aw + xU, \qquad u = (1-x)U.$$

Hence

$$\mathcal{I}(x,y) = \frac{\Gamma(2\epsilon)\Theta(0 \le x \le 1)\Theta(0 \le y \le 1-x)}{1-x} \int_0^{1/(1+a)} dw\, w^{\epsilon-1} U^{-1+\epsilon} [\Delta_x(w) - i0]^{-2\epsilon}, \tag{44}$$

with

$$U = 1 - (1+a)w.$$

The linear function $\Delta_x(w)$ can be written as

$$\Delta_x(w) = \Delta_0 + w\Delta_w, \tag{45}$$

where

$$\Delta_0 = \mu_l^2(x) - x(1-x)p^2, \tag{46}$$

$$\Delta_w = ((1+a)x(1-x) + b)p^2 - \mu_k^2(\lambda) + aM_l^2 - (1+a)\mu_l^2(x). \tag{47}$$

1.1.5 Map to Euler form

Set

$$t = (1+a)w, \tag{48}$$

so $t \in [0,1]$. Then

$$\mathcal{I}(x,y) = \frac{\Gamma(2\epsilon)\Theta}{1-x}(1+a)^{-\epsilon}(\Delta_0 - i0)^{-2\epsilon}\int_0^1 dt\, t^{\epsilon-1}(1-t)^{\epsilon-1}(1-zt)^{-2\epsilon}, \tag{49}$$

where

$$\Theta \equiv \Theta(0 \le x \le 1)\Theta(0 \le y \le 1-x), \qquad z = -\frac{\Delta_w}{(1+a)\Delta_0}. \tag{50}$$

Using Euler's integral,

$$\int_0^1 dt\, t^{\alpha-1}(1-t)^{\beta-1}(1-zt)^{-\rho} = B(\alpha,\beta)\, {}_2F_1(\rho,\alpha;\alpha+\beta;z),$$

with $\alpha = \beta = \epsilon$, $\rho = 2\epsilon$, gives

$$\mathcal{I}(x,y) = \frac{\Gamma(\epsilon)^2}{1-x}\Theta(1+a)^{-\epsilon}(\Delta_0 - i0)^{-2\epsilon}{}_2F_1(2\epsilon,\epsilon;2\epsilon;z). \tag{51}$$

Now use

$${}_2F_1(A,B;A;z) = (1-z)^{-B},$$

so that

$${}_2F_1(2\epsilon,\epsilon;2\epsilon;z) = (1-z)^{-\epsilon}.$$

This combines with the prefactors into a second scale Δ_1. With the present sign convention for $\Delta_x(w)$,

$$\Delta_1 = -\big[(1+a)\Delta_0 + \Delta_w\big]. \tag{52}$$

1.1.6 Final exact kernel

We obtain

$$\boxed{\mathcal{I}(x,y) = \frac{\Gamma(\epsilon)^2}{1-x}\Theta(0 \le x \le 1)\Theta(0 \le y \le 1-x)[\Delta_0 - i0]^{-\epsilon}[\Delta_1 - i0]^{-\epsilon}} \tag{53}$$

with

$$\Delta_0 = \mu_l^2(x) - x(1-x)p^2, \tag{54}$$

and

$$\Delta_1 = \mu_k^2(\lambda) - \lambda(1-\lambda)M_l^2 - \lambda(1-\lambda x)p^2, \qquad \lambda = \frac{y}{1-x}. \tag{55}$$

Equivalently,

$$\Delta_1 = \left(1 - \frac{y}{1-x}\right)m_k^2 + \frac{y}{1-x}M_k^2 - \frac{y}{1-x}\left(1 - \frac{y}{1-x}\right)M_l^2 - \frac{y}{1-x}\left(1 - \frac{xy}{1-x}\right)p^2. \tag{56}$$

1.2 Method 2: Consecutive Parametrization and Light-Cone Decomposition

1.2.1 Combining the two quadratic l propagators

Define

$$A = l^2 - m_l^2 + \mathrm{i}0, \qquad B = (l-p)^2 - M_l^2 + \mathrm{i}0. \tag{57}$$

We combine them by

$$\frac{1}{AB} = \int_0^1 \mathrm{d}u \, \frac{1}{[uA + (1-u)B]^2}. \tag{58}$$

A short calculation gives

$$uA + (1-u)B = \big(l - (1-u)p\big)^2 - \mu_l^2(u) + u(1-u)p^2 + \mathrm{i}0. \tag{59}$$

We therefore shift

$$L \equiv l - (1-u)p, \qquad l = L + (1-u)p, \qquad p - l = up - L. \tag{60}$$

The delta function becomes

$$\delta(x - l\cdot\eta) = \delta\big(L\cdot\eta - (x - 1 + u)\big). \tag{61}$$

Hence

$$\begin{aligned} \mathcal{I}(x,y) = \Gamma(\epsilon)\, \frac{\Theta(0 \le y \le 1-x)}{1-x} \int_0^1 \mathrm{d}u \int \frac{\mathrm{d}^d L}{\mathrm{i}\pi^{d/2}} \delta\big(L\cdot\eta - (x-1+u)\big) \\ \times \frac{[\mathcal{M}_u(L;x,y) - \mathrm{i}0]^{-\epsilon}}{[L^2 - \mu_l^2(u) + u(1-u)p^2 + \mathrm{i}0]^2}, \end{aligned} \tag{62}$$

where

$$\mathcal{M}_u(L;x,y) = \mu_k^2(\lambda) - \lambda(1-\lambda)(L^2 - 2uL\cdot p + u^2p^2) - \lambda(1-\lambda x)p^2. \tag{63}$$

1.2.2 The remaining light-cone decomposition

Now decompose

$$L = \xi p + \alpha\eta + L_\perp, \qquad L\cdot\eta = \xi, \qquad L\cdot p = \alpha + \xi p^2, \qquad L^2 = 2\xi\alpha + \xi^2p^2 + L_\perp^2. \tag{64}$$

The delta function fixes

$$\xi = x - 1 + u. \tag{65}$$

We introduce

$$a_u \equiv u - 1 + x. \tag{66}$$

Define also

$$X \equiv 1 - x, \qquad \kappa \equiv \lambda(1-\lambda) = \frac{y(1-x-y)}{(1-x)^2}. \tag{67}$$

Then the squared denominator is affine in α:

$$L^2 - \mu_l^2(u) + u(1-u)p^2 + \mathrm{i}0 = 2a_u\alpha + \Lambda_u(L_\perp;x) + \mathrm{i}0, \tag{68}$$

with

$$\Lambda_u(L_\perp;x) = a_u^2p^2 + L_\perp^2 - \mu_l^2(u) + u(1-u)p^2. \tag{69}$$

The noninteger factor is also affine in α:

$$\mathcal{M}_u(\alpha, L_\perp;x,y) = \beta_u(x,y)\alpha + \rho_u(L_\perp;u,x,y), \tag{70}$$

where

$$\beta_u(x,y) = 2\lambda(1-\lambda)(1-x) = \frac{2y(1-x-y)}{1-x}, \tag{71}$$

and

$$\begin{aligned}\rho_u(L_\perp; u, x, y) = \mu_k^2(\lambda) - \lambda(1-\lambda)\left[(x-1+u)(x-1-u)p^2 + L_\perp^2\right] \\ - \lambda(1-\lambda x)p^2.\end{aligned} \tag{72}$$

Consequently,

$$\mathcal{I}(x,y) = \Gamma(\epsilon)\,\frac{\Theta(0 \le y \le 1-x)}{1-x}\int_0^1 \mathrm{d}u \int \frac{\mathrm{d}\alpha\,\mathrm{d}^{d-2}L_\perp}{\mathrm{i}\pi^{d/2}}\,\frac{[\beta_u\alpha + \rho_u - \mathrm{i}0]^{-\epsilon}}{[2a_u\alpha + \Lambda_u + \mathrm{i}0]^2}. \tag{73}$$

1.2.3 Finite-ϵ evaluation of the α integral

Define

$$D_u(\alpha) = 2a_u\alpha + \Lambda_u + \mathrm{i}0, \qquad C_u(\alpha) = \beta_u\alpha + \rho_u - \mathrm{i}0. \tag{74}$$

We evaluate

$$J_\epsilon(u, L_\perp) \equiv \Gamma(\epsilon)\int_{-\infty}^{\infty} \mathrm{d}\alpha\,\frac{C_u(\alpha)^{-\epsilon}}{D_u(\alpha)^2}. \tag{75}$$

The two α-dependent factors are combined as

$$\frac{\Gamma(\epsilon)}{D_u^2 C_u^\epsilon} = \Gamma(2+\epsilon)\int_0^1 \mathrm{d}r\, r(1-r)^{\epsilon-1}\frac{1}{[rD_u + (1-r)C_u]^{2+\epsilon}}. \tag{76}$$

The denominator in (76) is affine in α:

$$rD_u + (1-r)C_u = P(r,u)\alpha + Q(r,u,L_\perp) + \mathrm{i}0_r, \tag{77}$$

where

$$P(r,u) = 2ra_u + (1-r)\beta_u, \tag{78}$$

and

$$Q(r,u,L_\perp) = r\Lambda_u + (1-r)\rho_u. \tag{79}$$

Distributional α integral Let

$$\mathcal{A}_\nu(P,Q) \equiv \int_{-\infty}^{\infty} \mathrm{d}\alpha\,[P\alpha + Q + \mathrm{i}0]^{-\nu} \tag{80}$$

For $P \neq 0$, the change of variables $t = P\alpha + Q$ gives a total derivative:

$$\int_{-\infty}^{\infty} \mathrm{d}t\,(t+\mathrm{i}0)^{-\nu} = -\frac{1}{\nu-1}\left[(t+\mathrm{i}0)^{1-\nu}\right]_{-\infty}^{+\infty} = 0. \tag{81}$$

Thus the α integral vanishes for $P \neq 0$. For $P = 0$, it is not an ordinary function but a distribution supported at $P = 0$. To evaluate the α integral we use the Schwinger representation. For $\mathrm{Re}\,\nu > 0$,

$$(x+\mathrm{i}0)^{-\nu} = \frac{e^{-\mathrm{i}\pi\nu/2}}{\Gamma(\nu)}\int_0^\infty \mathrm{d}s\, s^{\nu-1}e^{\mathrm{i}s(x+\mathrm{i}0)}. \tag{82}$$

This follows from the Euclidean Schwinger formula

$$A^{-\nu} = \frac{1}{\Gamma(\nu)}\int_0^\infty \mathrm{d}s\, s^{\nu-1}e^{-sA}, \qquad \mathrm{Re}\,A > 0, \tag{83}$$

by setting

$$A = -\mathrm{i}(x + \mathrm{i}0). \tag{84}$$

Moreover,

$$e^{-sA} = e^{\mathrm{i}s(x+\mathrm{i}0)}, \tag{85}$$

and the phase factor $e^{-\mathrm{i}\pi\nu/2}$ converts $[-\mathrm{i}(x + \mathrm{i}0)]^{-\nu}$ back to $(x + \mathrm{i}0)^{-\nu}$. In our case,

$$x = P\alpha + Q. \tag{86}$$

Therefore Eq. (82) gives

$$\begin{aligned}(P\alpha + Q + \mathrm{i}0)^{-\nu} &= \frac{e^{-\mathrm{i}\pi\nu/2}}{\Gamma(\nu)} \int_0^\infty \mathrm{d}s\, s^{\nu-1} e^{\mathrm{i}s(Q+\mathrm{i}0)} e^{\mathrm{i}sP\alpha} &(87)\\ &= 2\pi\delta(P)\frac{e^{-\mathrm{i}\pi\nu/2}}{\Gamma(\nu)} \int_0^\infty \mathrm{d}s\, s^{\nu-2} e^{\mathrm{i}s(Q+\mathrm{i}0)} &(88)\end{aligned}$$

where we used that the α dependence is now purely Fourier-like

$$\int_{-\infty}^{\infty} \mathrm{d}\alpha\, e^{\mathrm{i}sP\alpha} = 2\pi\delta(sP). \tag{89}$$

and that

$$\delta(sP) = \frac{1}{s}\delta(P), \qquad s > 0. \tag{90}$$

The remaining s integral has one lower power of s. Applying the same Schwinger identity with $\nu \to \nu - 1$ gives

$$\int_0^\infty \mathrm{d}s\, s^{\nu-2} e^{\mathrm{i}s(Q+\mathrm{i}0)} = \Gamma(\nu - 1) e^{\mathrm{i}\pi(\nu-1)/2} (Q + \mathrm{i}0)^{-(\nu-1)}. \tag{91}$$

Therefore

$$\mathcal{A}_\nu(P, Q) = 2\pi \frac{\Gamma(\nu - 1)}{\Gamma(\nu)} e^{-\mathrm{i}\pi\nu/2} e^{\mathrm{i}\pi(\nu-1)/2} \delta(P)(Q + \mathrm{i}0)^{-(\nu-1)}. \tag{92}$$

Since

$$\frac{\Gamma(\nu - 1)}{\Gamma(\nu)} = \frac{1}{\nu - 1}, \qquad e^{-\mathrm{i}\pi\nu/2} e^{\mathrm{i}\pi(\nu-1)/2} = e^{-\mathrm{i}\pi/2} = -\mathrm{i}, \tag{93}$$

we obtain

$$\boxed{\mathcal{A}_\nu(P, Q) = -\frac{2\pi\mathrm{i}}{\nu - 1}\delta(P)(Q + \mathrm{i}0)^{-(\nu-1)}.} \tag{94}$$

For the present integral, $\nu = 2 + \epsilon$, so

$$\boxed{\int_{-\infty}^{\infty} \mathrm{d}\alpha\, [P\alpha + Q + \mathrm{i}0]^{-2-\epsilon} = -\frac{2\pi\mathrm{i}}{1 + \epsilon}\delta(P)(Q + \mathrm{i}0)^{-1-\epsilon}.} \tag{95}$$

Collapse of the r integral and induced u support Applying Eq. (95) to Eq. (76) gives

$$J_\epsilon(u, L_\perp) = -\frac{2\pi\mathrm{i}}{1 + \epsilon}\Gamma(2 + \epsilon) \int_0^1 \mathrm{d}r\, r(1 - r)^{\epsilon-1} \delta\big(P(r, u)\big) \big[Q(r, u, L_\perp) + \mathrm{i}0_r\big]^{-1-\epsilon}. \tag{96}$$

Since

$$P(r, u) = \beta_u + r(2a_u - \beta_u), \tag{97}$$

the zero of P is

$$r_\star(u) = \frac{\beta_u}{\beta_u - 2a_u} = \frac{\beta_u}{\beta_u + 2(X - u)}. \tag{98}$$

Moreover,

$$\partial_r P(r,u) = 2a_u - \beta_u. \tag{99}$$

Therefore

$$\int_0^1 \mathrm{d}r\, f(r)\delta(P(r,u)) = \begin{cases} \dfrac{f(r_\star)}{|\beta_u - 2a_u|}, & 0 < r_\star < 1, \\ 0, & r_\star \notin (0,1). \end{cases} \tag{100}$$

The Feynman parameter r is integrated only over $0 \le r \le 1$, so the delta function contributes only when its zero lies in this interval. On the support $0 \le y \le 1-x$ one has $\beta_u \ge 0$. In the interior, $\beta_u > 0$, and the condition $0 < r_\star < 1$ is equivalent to

$$a_u < 0. \tag{101}$$

Since $a_u = u - X$, this gives

$$\boxed{0 < u < X = 1 - x.} \tag{102}$$

Thus the r-integration range induces the remaining support in u.

1.2.4 Current integral after the α and r integrations

Evaluating the integrand at $r_\star$ gives

$$\begin{aligned} \frac{r_\star(1-r_\star)^{\epsilon-1}}{|\partial_r P(r_\star,u)|} &= \frac{\dfrac{\beta_u}{\beta_u + 2(X-u)}\left[\dfrac{2(X-u)}{\beta_u + 2(X-u)}\right]^{\epsilon-1}}{\beta_u + 2(X-u)} \\ &= \frac{\beta_u[2(X-u)]^{\epsilon-1}}{[\beta_u + 2(X-u)]^{1+\epsilon}}. \end{aligned} \tag{103}$$

Furthermore,

$$\begin{aligned} Q(r_\star,u,L_\perp) &= r_\star\Lambda_u + (1-r_\star)\rho_u \\ &= \frac{\beta_u\Lambda_u + 2(X-u)\rho_u}{\beta_u + 2(X-u)} \\ &= \frac{\beta_u\Lambda_u - 2a_u\rho_u}{\beta_u + 2(X-u)}. \end{aligned} \tag{104}$$

Define

$$\mathcal{D}_u(L_\perp) \equiv \beta_u\Lambda_u - 2a_u\rho_u. \tag{105}$$

Then

$$[Q(r_\star,u,L_\perp) + \mathrm{i}0]^{-1-\epsilon} = [\beta_u + 2(X-u)]^{1+\epsilon}[\mathcal{D}_u(L_\perp) - \mathrm{i}0]^{-1-\epsilon}. \tag{106}$$

The factor $[\beta_u + 2(X-u)]^{1+\epsilon}$ cancels the denominator in Eq. (103). Therefore the net result after the r and α integrations is proportional to

$$\beta_u[2(X-u)]^{\epsilon-1}[\mathcal{D}_u(L_\perp) - \mathrm{i}0]^{-1-\epsilon}. \tag{107}$$

Thus the full integral at this stage is

$$\boxed{\mathcal{I}(x,y) = \frac{\Theta_{xy}}{X}\, 2\Gamma(1+\epsilon)\int_0^X \mathrm{d}u\,\beta_u[2(X-u)]^{\epsilon-1}\int \frac{\mathrm{d}^{d-2}L_\perp}{\pi^{(d-2)/2}}\big[\mathcal{D}_u(L_\perp) - \mathrm{i}0\big]^{-1-\epsilon}.} \tag{108}$$

The remaining tasks are the transverse $L_\perp$ integration and the final one-dimensional integration over u.

1.2.5 Transverse integration and one-dimensional representation

The transverse dependence of $\mathcal{D}_u$ is linear in $L_\perp^2$. Using $\beta_u = 2X\kappa$, one obtains

$$\mathcal{D}_u(L_\perp) = 2u\kappa L_\perp^2 + C_u, \tag{109}$$

where

$$\begin{aligned} C_u = & -2X\kappa\,\mu_l^2(u) - 2a_u\,\mu_k^2(\lambda) \\ & + 2\left\{X\kappa\big[a_u^2 + u(1-u)\big] + a_u\kappa\big[X^2 - u^2\big] + a_u\frac{y}{1-x}\left(1 - \frac{xy}{1-x}\right)\right\}p^2. \end{aligned} \tag{110}$$

The transverse integral yields with $n = d - 2 = 2 - 2\epsilon$

$$\int \frac{\mathrm{d}^n L_\perp}{\pi^{n/2}}\big[2u\kappa L_\perp^2 + C_u - \mathrm{i}0\big]^{-1-\epsilon} = \frac{\Gamma(2\epsilon)}{\Gamma(1+\epsilon)}(2u\kappa)^{-1+\epsilon}(C_u - \mathrm{i}0)^{-2\epsilon}. \tag{111}$$

Substituting Eq. (111) into Eq. (108), and using $\beta_u = 2X\kappa$, gives

$$\boxed{\mathcal{I}(x,y) = \Theta_{xy}\,\Gamma(2\epsilon)(4\kappa)^\epsilon \int_0^X \mathrm{d}u\, u^{-1+\epsilon}(X-u)^{-1+\epsilon}(C_u - \mathrm{i}0)^{-2\epsilon}.} \tag{112}$$

The endpoint values of C_u are

$$C_0 = 2X\Delta_1, \qquad C_X = -2X\kappa\Delta_0. \tag{113}$$

1.2.6 Exact matching of the finite-ϵ integral

The one-dimensional integral (112) is now mapped to the Euler form from Method 1. Use

$$w = \frac{X-u}{X-u+X\kappa}, \tag{114}$$

or equivalently

$$u = X - \frac{X\kappa w}{1-w}. \tag{115}$$

The endpoints map as

$$u = X \Longleftrightarrow w = 0, \qquad u = 0 \Longleftrightarrow w = \frac{1}{1+\kappa}. \tag{116}$$

Furthermore,

$$U = 1 - (1+\kappa)w. \tag{117}$$

Under this change of variables, Eq. (112) becomes

$$\boxed{\mathcal{I}_{\mathrm{M2}}(x,y) = \Gamma(2\epsilon)\frac{\Theta_{xy}}{1-x}\int_0^{1/(1+\kappa)} \mathrm{d}w\, w^{\epsilon-1}U^{-1+\epsilon}\big[\Delta_x(w) - \mathrm{i}0\big]^{-2\epsilon}.} \tag{118}$$

This is precisely the Method-1 representation Eq. (44). Therefore

$$\boxed{\mathcal{I}_{\mathrm{M2}}(x,y;\epsilon) = \mathcal{I}_{\mathrm{M1}}(x,y;\epsilon)} \tag{119}$$

as finite-ϵ integrals. Hence this method also yields the exact compact result

$$\boxed{\mathcal{I}_{\mathrm{M2}}(x,y) = \frac{\Theta_{xy}}{1-x}\Gamma(\epsilon)^2(\Delta_0 - \mathrm{i}0)^{-\epsilon}(\Delta_1 - \mathrm{i}0)^{-\epsilon}.} \tag{120}$$

1.3 Use as universal two-loop collinear block

For a generic hard subgraph $G(x, y)$,

$$I_G^{(2)} = \int_0^1 dx \int_0^{1-x} dy\, \mathcal{I}(x, y) G(x, y). \tag{121}$$

This is the direct two-loop analogue of the one-loop convolution $I_G = \int dx\, I_c(x) G(x)$.